Supplementary Material: Molecular Docking: A Machine-Checked Theory of Exact Resolution, Complexity, and Thermodynamic Cost

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AC1	ClaimClosure.AtomicCircuitExports.AC1 paper4/DecisionQuotient/ClaimClosure.lean	AC3	ClaimClosure.AtomicCircuitExports.AC3 paper4/DecisionQuotient/ClaimClosure.lean
AC4	ClaimClosure.AtomicCircuitExports.AC4 paper4/DecisionQuotient/ClaimClosure.lean	BA1	Physics.BoundedAcquisition.BoundedRegion paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA2	Physics.BoundedAcquisition.acquisition_rate_bound paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA3	Physics.BoundedAcquisition.acquisitions_are_transitions paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA4	Physics.BoundedAcquisition.one_bit_per_transition paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA5	Physics.BoundedAcquisition.resolution_reads_sufficient paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA6	Physics.BoundedAcquisition.srank_le_resolution_bits paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA7	Physics.BoundedAcquisition.energy_ge_srank_cost paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
BA8	Physics.BoundedAcquisition.srank_one_energy_minimum paper4/DecisionQuotient/Physics/BoundedAcquisition.lean	BA10	Physics.BoundedAcquisition.counting_gap_theorem paper4/DecisionQuotient/Physics/BoundedAcquisition.lean
CV7	Physics.Conversation.clamp_projection_eq_iff_same_clamped_bit paper4/DecisionQuotient/Physics/Conversation.lean	CV8	Physics.Conversation.clampDecisionEvent_iff_bitOps_pos paper4/DecisionQuotient/Physics/Conversation.lean
CV9	Physics.Conversation.clamp_event_implies_positive_energy paper4/DecisionQuotient/Physics/Conversation.lean	DT22	Physics.DecisionTime.substrate_step_realizes_decision_event paper4/DecisionQuotient/Physics/DecisionTime.lean
DT23	Physics.DecisionTime.substrate_step_is_time_unit paper4/DecisionQuotient/Physics/DecisionTime.lean	DT24	Physics.DecisionTime.time_unit_law_substrate_invariant paper4/DecisionQuotient/Physics/DecisionTime.lean
EI1	ThermodynamicLift.energy_ge_kbt_nat_entropy paper4/DecisionQuotient/ThermodynamicLift.lean	IT3	DecisionQuotient.quotientEntropy_le_srank_binary paper4/DecisionQuotient/Information.lean
IT4	DecisionQuotient.numOptClasses_le_pow_srank_binary paper4/DecisionQuotient/Information.lean	L17	Leverage.compose_dof Leverage/Foundations.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
L19	Leverage.composition_dof_additive Leverage/Theorems.lean	L43	dof_eq_srank Leverage/BridgeToDQ.lean
L44	dof_one_iff_max_leverage Leverage/Foundations.lean	L45	england_replication_inequality Leverage/BridgeToDQ.lean
L46	incoherent_srank_gt_one Leverage/BridgeToDQ.lean	L47	max_coherence_forces_tractability Leverage/BridgeToDQ.lean
L49	srank_energy_lower_bound Leverage/BridgeToDQ.lean	L51	ssot_srank_one Leverage/BridgeToDQ.lean
L52	succ_le_two_pow Leverage/BridgeToDQ.lean	L53	sufficiency_conp_hard
L54	thermodynamic_selection Leverage/BridgeToDQ.lean	L55	thermodynamic_selection_unconditional Leverage/BridgeToDQ.lean
L56	tractable_bounded_core paper4/DecisionQuotient/ClaimClosure.lean	L57	Leverage.ColumnComplexityBridge. SharedCodewordCount_eq_TieBrokenRelationMoment Leverage/ColumnComplexityBridge.lean
L58	Leverage.ColumnComplexityBridge. zeroIdentityDebt_tieBrokenArgmin_of_uniform_ argmin_relation_bound Leverage/ColumnComplexityBridge.lean	L60	Leverage.rattle_constraintObservations_card Leverage/BridgeToDQ.lean
L61	Leverage.rattle_energy_lower_bound Leverage/BridgeToDQ.lean	L62	Leverage.rattle_srank_eq_effectiveDOF Leverage/BridgeToDQ.lean
L63	Leverage.exactSufficiency_conp_core Leverage/DockingTheoryBridge.lean	L64	Leverage.exactSufficiency_hardFamily_srank_eq_ n Leverage/DockingTheoryBridge.lean
L65	Leverage.molecularDocking_boundedPocket_srank_ bound Leverage/DockingTheoryBridge.lean	L66	Leverage.molecularDocking_srank_bound Leverage/DockingTheoryBridge.lean
L67	Leverage.sampledDocking_exactCoarse_opt_agree_ of_gap Leverage/DockingTheoryBridge.lean	L68	Leverage.sampledDocking_insideCutoff_ sufficient Leverage/DockingTheoryBridge.lean
L69	Leverage.exactSufficiency_checkerBudget_ge_ witnessBudget Leverage/DockingTheoryBridge.lean	L70	Leverage.exactSufficiency_checkingTime_ge_ witnessBudget Leverage/DockingTheoryBridge.lean
L71	Leverage.exactSufficiency_noSoundChecker_ below_witnessBudget Leverage/DockingTheoryBridge.lean	L72	Leverage.exactTopK_subset_ambiguityBand Leverage/DockingTheoryBridge.lean
L73	Leverage.exactVsCutoffCoulomb_opt_invariance Leverage/DockingTheoryBridge.lean	L74	Leverage.exactVsCutoffCoulomb_uniformApprox Leverage/DockingTheoryBridge.lean
L75	Leverage.exactVsCutoffLJ_opt_invariance Leverage/DockingTheoryBridge.lean	L76	Leverage.fisherMatrix_rank_eq_srank Leverage/DockingTheoryBridge.lean
L77	Leverage.surjectiveAbstraction_factors_or_ erases Leverage/DockingTheoryBridge.lean	L78	Leverage.surjectiveAbstraction_ withFeasibleCollapseMap_factors Leverage/DockingTheoryBridge.lean
L79	Leverage.topKPreserved_of_boundaryGap Leverage/DockingTheoryBridge.lean	L80	Leverage.totalFisher_eq_srank Leverage/DockingTheoryBridge.lean
L81	Leverage.ewaldFourier_positive Leverage/DockingTheoryBridge.lean	L82	Leverage.ewaldRealSpace_exponentialDecay Leverage/DockingTheoryBridge.lean
L83	Leverage.lennardJones_gradient Leverage/DockingTheoryBridge.lean	L84	Leverage.multiplicativeSeparable_ emptySufficient Leverage/DockingTheoryBridge.lean
L85	Leverage.strictGlobalDominance_emptySufficient Leverage/DockingTheoryBridge.lean	L86	Leverage.velocityVerlet_preservesVolume Leverage/DockingTheoryBridge.lean
L87	Leverage.boundedActions_polynomialTime Leverage/DockingTheoryBridge.lean	L88	Leverage.gridMDState_sufficient_erase_ irrelevant Leverage/DockingTheoryBridge.lean
L89	Leverage.largeCutoff_implies_bounded Leverage/DockingTheoryBridge.lean	L90	Leverage.lipschitzUtilityApprox_implies_ resolutionControlled Leverage/DockingTheoryBridge.lean
L91	Leverage.resolutionControlledApprox_implies_ uniformApprox Leverage/DockingTheoryBridge.lean	L92	Leverage.sampledDocking_ finiteUniformErrorRadius_witnesses Leverage/DockingTheoryBridge.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
L93	Leverage.sufficiency_poly_bounded_actions Leverage/DockingTheoryBridge.lean	L94	Leverage.topKMargin_certificate_sound Leverage/DockingTheoryBridge.lean
L95	Leverage.boundedPotential_largeCutoff_srank_bound Leverage/DockingTheoryBridge.lean	L96	Leverage.exactCoulomb_largeCutoff_srank_bound Leverage/DockingTheoryBridge.lean
L97	Leverage.exactLJ_largeCutoff_srank_bound Leverage/DockingTheoryBridge.lean	L98	Leverage.exactRealEwald_largeCutoff_srank_bound Leverage/DockingTheoryBridge.lean
L99	Leverage.resolutionControlled_gap_implies_opt_invariance Leverage/DockingTheoryBridge.lean	L100	Leverage.exactLJ_distanceError_gap_implies_opt_invariance Leverage/DockingTheoryBridge.lean
L101	Leverage.exactLJ_uniformApprox_of_distanceError_and_shellGradBound Leverage/DockingTheoryBridge.lean	L102	Leverage.lennardJones_value_diff_le_of_shellGradBound Leverage/DockingTheoryBridge.lean
L103	Leverage.abs_ljGradExpr_le_shellBound Leverage/DockingTheoryBridge.lean	L104	Leverage.boundedPotential_largeCutoff_sampledSrank_bound Leverage/DockingTheoryBridge.lean
L105	Leverage.abs_ljSecondDerivExpr_le_shellBound Leverage/DockingTheoryBridge.lean	L106	Leverage.lennardJones_grad_diff_le_of_shellSecondDerivBound Leverage/DockingTheoryBridge.lean
L107	Leverage.exactLJ_quadraticDiscretizationError_of_distanceError_and_shellSecondDerivShellBound Leverage/DockingTheoryBridge.lean	L108	Leverage.lennardJones_taylor_remainder_le_of_shellSecondDerivShellBound Leverage/DockingTheoryBridge.lean
L109	Leverage.molecularDocking_above_ground_of_srank_gt_one Leverage/DockingTheoryBridge.lean	L110	Leverage.molecularDocking_landauer_entropy_lower_bound Leverage/DockingTheoryBridge.lean
L111	Leverage.molecularDocking_srank_energy_lower_bound Leverage/DockingTheoryBridge.lean	L112	Leverage.sampledDocking_energy_le_boundedAcquisitions_of_insideCutoff_and_budget Leverage/DockingTheoryBridge.lean
L113	Leverage.sampledDocking_flatExact_natEntropy_le_srank_log_gridArity Leverage/DockingTheoryBridge.lean	L114	Leverage.sampledDocking_flatBinaryEncoded_landauer_entropy_lower_bound Leverage/DockingTheoryBridge.lean
L115	Leverage.sampledDocking_flatActionPinned_srank_le_srank Leverage/DockingTheoryBridge.lean	L116	Leverage.sampledDocking_flatActionPinned_strictOpt Leverage/DockingTheoryBridge.lean
L117	Leverage.sampledDocking_flatActionPinned_uniformStrictUtilityGap_of_card_gt_one Leverage/DockingTheoryBridge.lean	ORA1	oracle_arbitrary paper2/Ssot/Coherence.lean
PH26	PhysicalComplexity.no_collapse_of_bounded_budget_pos_cost_exp_lb paper4/DecisionQuotient/Physics/PhysicalHardness.lean	QT1	DecisionProblem.quotient_is_coarsest paper4/DecisionQuotient/Quotient.lean
QT2	DecisionProblem.quotientMap_preservesOpt paper4/DecisionQuotient/Quotient.lean	QT3	DecisionProblem.quotient_represents_opt_equiv paper4/DecisionQuotient/Quotient.lean
QT7	DecisionProblem.quotient_has_unique_factorization paper4/DecisionQuotient/Quotient.lean	SE1	ClaimClosure.SE1 paper4/DecisionQuotient/ClaimClosure.lean
SE2	ClaimClosure.SE2 paper4/DecisionQuotient/ClaimClosure.lean	SE3	ClaimClosure.SE3 paper4/DecisionQuotient/ClaimClosure.lean
SE4	ClaimClosure.SE4 paper4/DecisionQuotient/ClaimClosure.lean	SE5	ClaimClosure.SE5 paper4/DecisionQuotient/ClaimClosure.lean
SE6	ClaimClosure.SE6 paper4/DecisionQuotient/ClaimClosure.lean	W1	Physics.single_future_zero_cost paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean
W2	Physics.transportCost_pos_of_offDiag paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean	W3	Physics.integrity_is_centroid paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean
W4	Physics.wasserstein_bridge paper4/DecisionQuotient/Physics/WassersteinIntegrity.lean	WM4	Physics.WolpertMismatch.mismatchNatLowerBound_pos_of_exists_ne paper4/DecisionQuotient/Physics/WolpertMismatch.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
WM6	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_distribution_mismatch paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP2	Physics.WolpertDecomposition.landauer_floor_plus_decomposition_lower_bound paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WP6	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_either_cited_component paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP7	Physics.WolpertDecomposition.landauer_floor_plus_structural_resource_lower_bound paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WP8	Physics.WolpertDecomposition.energy_lower_bound_increases_by_structural_resource paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WP9	Physics.WolpertDecomposition.physical_grounding_bundle_with_wolpert_decomposition paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WR6	Physics.WolpertResidual.discreteResidualNatLowerBound_pos_of_asymmetry_or_oneway paper4/DecisionQuotient/Physics/WolpertResidual.lean	WR7	Physics.WolpertDecomposition.stopping_time_residual_of_discrete_edge_split paper4/DecisionQuotient/Physics/WolpertDecomposition.lean
WR10	Physics.WolpertDecomposition.effective_model_strictly_exceeds_landauer_of_finite_discrete_witness paper4/DecisionQuotient/Physics/WolpertDecomposition.lean	WR11	Physics.WolpertResidual.binaryEncodedResidualNatLowerBound_eq_one paper4/DecisionQuotient/Physics/WolpertResidual.lean
WR12	Physics.WolpertDecomposition.effective_model_ge_landauer_plus_one_of_binary_encoded_residual_example paper4/DecisionQuotient/Physics/WolpertDecomposition.lean		

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 4.7: Bounded-Pocket Low-Rank Regime	L65
Corollary 5.11: Decision-Entropy Bound from Spacetime and Energy Budget	IT3, BA1, BA2, BA5, BA6, EI1, L43
Corollary 4.5: Checking Time Lower Bound	L70
Corollary 5.12: Independent Composition Budget Law	L17, L19, BA1, BA2, BA7, BA5, BA6, L43
Corollary 5.30: RATTLE Holonomic-Constraint Landauer Floor	L60, L61, L62
Corollary 5.4: Unique Minimum-Cost Regime	BA8, L54, L55
Corollary 4.4: No Sound Checker Below Witness Budget	L71
Corollary 3.7: Higher-Rank Regime	L46
Corollary 3.6: Rank-One Regime	L51
Proposition 5.14: Two-Level Atomic Realization	AC1, AC3, AC4
Proposition 5.26: Repeated Binary Mismatch Work Law	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.25: Binary Mismatch Strengthens the Energy–Information Coefficient	IT3, BA5, BA6, WP2, WM4, L43
Proposition 5.24: Explicit Binary Mismatch Example	BA5, BA6, WP2, WM4, L43
Proposition 5.19: Repeated Residual-Example Work Law	IT3, BA5, BA6, WR12, WR11, L43

Paper claim	Lean handle
Proposition 5.18: Explicit Two-State Residual Example	IT3, BA5, BA6, WR12, WR11, L43
Proposition 2.5: Bounded Region	BA1
Proposition 5.22: Canonical Wolpert Grounding Bundle	BA5, BA6, WP9, L43
Definition 2.3: Bounded Decision System	L17, L19
Proposition 5.20: Unified Energy–Information Hierarchy	IT3, BA5, BA6, WR12, WP2, WM4, WR11, L43
Proposition 3.11: Finite Compression-Relation Bridge	L57, L58
Proposition 5.17: Finite Discrete Residual Witness	WR10, WR7, WR6
Proposition 5.27: Positive Heat and Bounded Lifetime	SE1, SE2, SE3, SE4, SE5
Proposition 5.28: Finite Lifetime Throughput Bound	SE5, IT3, L43
Definition 2.13: Canonical Decision Problem	QT2, QT7, QT1, QT3
Proposition 5.29: Speed-Heat Tradeoff	SE6
Proposition 5.23: Strict Canonical Energy Above the Landauer Floor	BA5, BA6, WP6, L43
Proposition 5.16: Theorem-Level Strict Overhead Branches	BA5, BA6, WM6, WP6, WR10, L43
Proposition 5.21: Structural-Resource Overhead	BA5, BA6, WP8, WP7, L43
Proposition 5.15: Substrate Step is Unit Interface Time	DT23, DT22, DT24
Proposition 5.13: Threshold Channel Realization	CV8, CV9, CV7
Theorem 3.1: Surjective Abstractions Either Factor or Erase	L77
Theorem 4.11: Action-Pinned Sampled Docking Has a Uniform Positive Strict Gap	L115, L116, L117
Theorem 2.6: Bounded Acquisition Rate	BA1, BA2
Theorem 4.33: Bounded-Action Exact Checking Is Polynomial-Time Decidable	L87, L93
Theorem 4.28: Bounded Potential Plus Large Cutoff Gives Sampled Structural-Rank Bound	L104
Theorem 4.27: Bounded Potential Plus Large Cutoff Gives Structural-Rank Bound	L95
Theorem 5.10: Decision-Class Bound from Spacetime and Energy Budget	IT4, IT3, BA1, BA2, BA5, BA6, EI1, L43
Theorem 4.3: Checker Budget Lower Bound	L69
Theorem 6.1: Coherent Single-Source Regime	ORA1
Theorem 4.37: Cutoff Coulomb Winner Preservation	L73
Theorem 4.36: Cutoff Coulomb Uniform Approximation	L74
Theorem 4.38: Exact Coulomb Tail Control Gives Structural-Rank Bound	L96
Theorem 2.4: Counting Gap	BA10
Theorem 2.7: Discrete Acquisition	BA3
Theorem 3.5: DOF–Structural-Rank Identity	L43
Theorem 5.3: Energy–Information Duality	IT3, EI1, L43
Theorem 5.1: Rank Controls Exact-Resolution Cost	BA7, BA6, L43
Theorem 6.8: Finite Replication Entropy Gap	L45

Paper claim	**Lean handle**
Theorem 3.10: Decision-Entropy Bound	`IT3`, `L43`
Theorem 4.43: Ewald Reciprocal Core Is Positive	`L81`
Theorem 4.42: Ewald Real-Space Exponential Decay	`L82`
Theorem 4.39: Exact Real-Space Ewald Tail Control Gives Structural-Rank Bound	`L98`
Theorem 4.1: General Hardness Core for Exact Sufficiency	`L63`
Theorem 3.2: Feasible Collapse Maps Force Quotient Factorization	`L78`
Theorem 6.9: Finite-Budget No-Collapse	`PH26`
Theorem 4.29: Finite Sampled Docking Has a Canonical Uniform Error Radius	`L92`
Theorem 3.4: Fisher-Matrix Rank Equals Structural Rank	`L76`
Theorem 3.3: Total Fisher Information Equals Structural Rank	`L80`
Theorem 6.7: Convergence	`L43`, `L44`, `L47`, `L55`
Theorem 4.30: Finite Grid Exact-Docking Entropy Is Controlled by Structural Rank	`L113`
Theorem 4.32: Grid Irrelevance Erasure Is Sound	`L88`
Theorem 4.2: Maximal Structural Rank in the Hard Family	`L64`
Theorem 4.18: Large Cutoff Implies Docking Cutoff Boundedness	`L89`
Theorem 4.15: Lipschitz Grid Error Gives Resolution-Controlled Approximation	`L90`
Theorem 4.34: Cutoff Lennard-Jones Winner Preservation	`L75`
Theorem 4.40: Closed-Form Lennard-Jones Gradient	`L83`
Theorem 4.19: Explicit Lennard-Jones Shell Derivative Envelope	`L103`
Theorem 4.26: Lennard-Jones Shell Gradient Bound Preserves Exact Winners	`L100`
Theorem 4.23: Lennard-Jones Shell Gradient Bound Gives Score Stability	`L102`
Theorem 4.21: Lennard-Jones Hessian Envelope Makes the Gradient Lipschitz	`L106`
Theorem 4.25: First-Order-Corrected Lennard-Jones Discretization Error Is Quadratic	`L107`
Theorem 4.22: Lennard-Jones Hessian Envelope Gives a Quadratic Taylor Remainder	`L108`
Theorem 4.20: Explicit Lennard-Jones Shell Second-Derivative Envelope	`L105`
Theorem 4.24: Lennard-Jones Shell Gradient Bound Gives Uniform Exact/Grid Approximation	`L101`
Theorem 4.35: Exact Lennard-Jones Tail Control Gives Structural-Rank Bound	`L97`
Theorem 5.7: Higher-Rank Exact Docking Lies Above the Ground State	`L109`
Theorem 5.6: Binary-Encoded Finite Grid Docking Carries Landauer Cost	`L114`
Theorem 5.5: Exact Docking Structural Rank Controls Resolution Cost	`L111`
Theorem 3.8: Minimum Physical Bit Operations	`BA5`, `BA6`, `L43`
Theorem 4.6: Cutoff-Local Structural-Rank Bound for Exact Docking	`L66`
Theorem 4.9: Multiplicative Separability Eliminates Conformer Dependence	`L84`
Theorem 3.9: Decision-Class Bound	`IT4`, `L43`

Paper claim	Lean handle
Theorem 2.8: One Transition, One Bit	BA4
Theorem 6.3: Rank Identification	L43, L46, L51
Theorem 5.2: Rank-One Ground State	BA8, L54, L55
Theorem 4.17: Resolution-Controlled Approximation Preserves Exact Winners	L99
Theorem 4.16: Resolution-Controlled Approximation Gives Uniform Utility Approximation	L91
Theorem 2.9: Resolution Requires a Sufficient Coordinate Set	BA5
Theorem 4.10: Sampled Exact-Coarse Winner Preservation	L67
Theorem 5.8: Inside-Cutoff Sampled Exact Resolution Gives a Concrete Bounded-Region Energy Floor	L112
Theorem 4.12: Inside-Cutoff Sufficiency for Sampled Docking	L68
Theorem 4.8: Strict Dominance Makes Coordinates Irrelevant	L85
Theorem 6.5: Thermodynamic Selection	BA8, L49, L54, L55
Theorem 5.9: Exact-Resolution Time Lower Bound	BA1, BA2, BA5, BA6, L43
Theorem 4.14: Exact Top-k Ambiguity-Band Containment	L72
Theorem 4.13: Top-k Preservation Under Boundary Gap	L79
Theorem 4.31: Top-k Margin Certificates Are Sound	L94
Theorem 6.4: Tractable Sufficiency at Rank One	L47, L53, L56
Theorem 4.41: Velocity-Verlet Preserves Phase-Space Volume	L86

Auto summary: mapped 99/99 (full=99, derived=0, unmapped=0).

3 Proof Hardness Index

Paper handle	Hardness profile	Regime tags	Lean support
cor: bounded-pocket-regime	unspecified	-	L65
cor: budget-entropy-bound	unspecified	-	IT3, BA1, BA2, BA5, BA6, EI1, L43
cor: checking-time-lower-bound	unspecified	-	L70
cor: composition-budget-law	unspecified	-	L17, L19, BA1, BA2, BA7, BA5, BA6, L43
cor: holonomic-landauer-floor	unspecified	-	L60, L61, L62
cor: minimum-cost-regime	unspecified	-	BA8, L54, L55
cor: no-sound-checker-below-budget	unspecified	-	L71
cor:rank-above-one	unspecified	-	L46
cor:rank-one	unspecified	-	L51
prop: atomic-realization	unspecified	-	AC1, AC3, AC4
prop: binary-mismatch-cumulative-work	unspecified	-	IT3, BA5, BA6, WP2, WM4, L43
prop: binary-mismatch-energy-information	unspecified	-	IT3, BA5, BA6, WP2, WM4, L43

prop:binary-mismatch-example	unspecified	-	BA5, BA6, WP2, WM4, L43
prop:binary-residual-cumulative-work	unspecified	-	IT3, BA5, BA6, WR12, WR11, L43
prop:binary-residual-example	unspecified	-	IT3, BA5, BA6, WR12, WR11, L43
prop:bounded-region	unspecified	-	BA1
prop:canonical-wolpert-bundle	unspecified	-	BA5, BA6, WP9, L43
prop:dof-additive	unspecified	-	L17, L19
prop:ei-hierarchy	unspecified	-	IT3, BA5, BA6, WR12, WP2, WM4, WR11, L43
prop:finite-compression-bridge	unspecified	-	L57, L58
prop:finite-discrete-residual	unspecified	-	WR10, WR7, WR6
prop:finite-lifetime	unspecified	-	SE1, SE2, SE3, SE4, SE5
prop:lifetime-throughput	unspecified	-	SE5, IT3, L43
prop:optimizer-quotient	unspecified	-	QT2, QT7, QT1, QT3
prop:speed-heat-tradeoff	unspecified	-	SE6
prop:strict-canonical-energy	unspecified	-	BA5, BA6, WP6, L43
prop:strict-overhead	unspecified	-	BA5, BA6, WM6, WP6, WR10, L43
prop:structural-resource-overhead	unspecified	-	BA5, BA6, WP8, WP7, L43
prop:substrate-time-law	unspecified	-	DT23, DT22, DT24
prop:threshold-channel	unspecified	-	CV8, CV9, CV7
thm:abstraction-factors-or-erases	unspecified	-	L77
thm:action-pinned-uniform-gap	unspecified	-	L115, L116, L117
thm:bounded-acquisition	unspecified	-	BA1, BA2
thm:bounded-actions-poly	unspecified	-	L87, L93
thm:bounded-potential-large-cutoff-sampled-srank	unspecified	-	L104
thm:bounded-potential-large-cutoff-srank	unspecified	-	L95
thm:budget-class-bound	unspecified	-	IT4, IT3, BA1, BA2, BA5, BA6, EI1, L43
thm:checker-budget-lower-bound	unspecified	-	L69
thm:coherent-single-source	unspecified	-	ORA1
thm:coulomb-cutoff-invariance	unspecified	-	L73
thm:coulomb-cutoff-uniform-approx	unspecified	-	L74
thm:coulomb-tail-srank	unspecified	-	L96
thm:counting-gap	unspecified	-	BA10
thm:discrete-acquisition	unspecified	-	BA3
thm:dof-srank	unspecified	-	L43
thm:energy-entropy	unspecified	-	IT3, EI1, L43
thm:energy-rank	unspecified	-	BA7, BA6, L43
thm:england	unspecified	-	L45
thm:entropy-bound	unspecified	-	IT3, L43
thm:ewald-fourier-positive	unspecified	-	L81
thm:ewald-real-space-decay	unspecified	-	L82
thm:ewald-tail-srank	unspecified	-	L98

thm: exact-sufficiency-hardness-core	unspecified	-	L63
thm: feasible-collapse-factors	unspecified	-	L78
thm: finite-budget-no-collapse	unspecified	-	PH26
thm: finite-uniform-error-radius	unspecified	-	L92
thm: fisher-rank-srank	unspecified	-	L76
thm:fisher-sum-srank	unspecified	-	L80
thm:five-way	unspecified	-	L43, L44, L47, L55
thm: grid-docking-entropy	unspecified	-	L113
thm: grid-erase-irrelevant	unspecified	-	L88
thm: hard-family-srank	unspecified	-	L64
thm: large-cutoff-bounded	unspecified	-	L89
thm: lipschitz-grid-approx	unspecified	-	L90
thm: lj-cutoff-invariance	unspecified	-	L75
thm:lj-gradient	unspecified	-	L83
thm: lj-shell-derivative-envelope	unspecified	-	L103
thm: lj-shell-gap-invariance	unspecified	-	L100
thm: lj-shell-grad-stability	unspecified	-	L102
thm: lj-shell-hessian-lipschitz	unspecified	-	L106
thm: lj-shell-quadratic-discretization	unspecified	-	L107
thm: lj-shell-quadratic-remainder	unspecified	-	L108
thm: lj-shell-second-derivative-envelope	unspecified	-	L105
thm: lj-shell-uniform-approx	unspecified	-	L101
thm:lj-tail-srank	unspecified	-	L97
thm:md-above-ground	unspecified	-	L109
thm: md-energy-entropy	unspecified	-	L114
thm:md-energy-rank	unspecified	-	L111
thm: min-bit-operations	unspecified	-	BA5, BA6, L43
thm: molecular-docking-srank-bound	unspecified	-	L66
thm: multiplicative-separable-empty-sufficient	unspecified	-	L84
thm:numopt-bound	unspecified	-	IT4, L43
thm: one-transition-one-bit	unspecified	-	BA4
thm: rank-identification	unspecified	-	L43, L46, L51
thm:rank-one-ground	unspecified	-	BA8, L54, L55
thm: resolution-controlled-gap-invariance	unspecified	-	L99
thm: resolution-controlled-uniform	unspecified	-	L91
thm: resolution-sufficient	unspecified	-	BA5
thm: sampled-docking-gap	unspecified	-	L67
thm: sampled-inside-cutoff-budget-energy	unspecified	-	L112

thm: sampled-inside-cutoff-sufficient	unspecified	-	L68
thm: strict-dominance-empty-sufficient	unspecified	-	L85
thm: thermodynamic-selection	unspecified	-	BA8, L49, L54, L55
thm:time-lower-bound	unspecified	-	BA1, BA2, BA5, BA6, L43
thm: topk-ambiguity-band	unspecified	-	L72
thm: topk-boundary-gap	unspecified	-	L79
thm: topk-certificate-sound	unspecified	-	L94
thm: tractable-rank-one	unspecified	-	L47, L53, L56
thm: velocity-verlet-volume	unspecified	-	L86

Auto summary: indexed 99 claims by hardness profile (unspecified=99).

4 Assumption Ledger

4.1 Assumption Ledger (Auto)

Source files.

- `Leverage/BridgeToDQ.lean`
- `Leverage/FiveWayEquivalence.lean`
- `Leverage/Physical.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

First-principles Bayes derivation handles.

- (No Bayes-derivation theorem handles found.)

Compact Bayes handle IDs.

- (No compact Bayes alias IDs found.)